

ELI validation rules (“ELI Shapes”)

Abstract : this document describes the structure, content and usage of the ELI validation rules.

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What are the ELI validation rules ?

The ELI validation rules are a set of rules that enable a system to assess the consistency of some ELI metadata against a set of rules. Typically, it enables to assess the consistency of ELI metadata of one act encoded inside a webpage.

The typical errors that the validation rules enable to find are:

- An ELI property that is expressed on a LegalExpression while it is expected only on a LegalResource
- A property unknown in the ELI ontology
- A property that has a literal (string) value while a URI is expected
- A mandatory property that is missing
- etc...

Besides strict conformance checking, the validation rules also encode some recommended rules that, while not strictly mandatory in terms of the ELI ontology, are recommended and recognized best practices.

A different set of the ELI validation rules exist for each version of the ELI ontology (1.3, 1.4, 1.5).

The rules are expressed in the [SHACL language](#), and they can be used with a SHACL conformance validator.

Structure of ELI validation rules

Overview of the rules

For each version of the ELI ontology, 3 rule files are provided:

- “core” rules : the core rules encode the rules that must be obeyed when publishing ELI metadata in a webpage.
- “ranges” rules : The “ranges” rules contain a set of additional range constraints “on top of the core rules.
- “recommended” rules : this file contains a set of additional consistency rules that are not strictly speaking part of the ELI ontology but that are best practices, in particular regarding the Pillar 4 expectations.

“Core” rules

The scope of the core rules is limited to what is expected in a webpage. Hence it does not check for proper range of properties, as, in a webpage, the types of property values is unknown (e.g. when asserting “Act_A eli:changes Act_B”, since the type of “Act_B” is unknown, we cannot check if its type is consistent with what is expected for predicate “eli:changes”).

The scope of the core rules is what is in the ELI ontology, nothing more. In particular it checks for the mandatory properties documented in the ontology (eli:type_document, eli:title, eli:format, eli:language + the FRBR properties eli:realizes/eli:is_realized_by or eli:embodies/eli:is_embodied_by).

“Core” rules technical SHACL implementation

The rules are organized as follows:

1. SHACL Node Shapes checks that each node is an IRI and, where necessary, checks that the string “/eli/” is part of the IRI.
 - These Node Shapes have implicit targets that are the instances of each ELI ontology class.
2. Property Shapes are declared on each Node Shape, at the corresponding level of abstraction (e.g. eli:Work for properties expected on Works, eli:LegalResource for properties expected on LegalResources). Each property shapes contain:
 - the expected value kind (IRI vs. Literal)
 - the cardinality expectation (min / max)
 - the expected datatype, when applicable
 - the expected set of possible values, when applicable
3. Additional Node Shapes are added to check the domains of the property, i.e. to check that each property is asserted on an object of the correct type.
 - There is one such Node Shape for each property
 - These Node Shape use a “target subjects of” of the corresponding property
 - They use an sh:class constraint to check that the subject of the property has the expected type.
4. There are additional “business rules”, currently only one, implemented as a query, that checks that for each (individual) Legal Resource, at least one eli:Format is reachable that would qualify for the Pillar 4 full-text retrieval.

“Ranges” rules

This “ranges” rules adds range constraints on top of the “Core” rules. *This file cannot be used alone, it must be used in combination with the core rules.*

This file is usable only in cases where the system needs to validate the consistency of a complete knowledge graph, and not only the consistency of what has been published in a single webpage.

“Ranges” rules technical SHACL implementation

This file refers to the Property Shapes of the core files and adds only value class constraints, e.g.

“`elish:Work_is_realized_by sh:class eli:Expression`”.

It is thus important to ensure that the IRI of the Property Shapes are consistent between the “core” rules and the “ranges” rules.

“Recommended” rules

Recommended rules can be used in addition to the core rules and ranges rules. It is not mandatory to use them.

The recommended rules contain:

- a set of extra domain checks : these rules will verify that, for each property of the ELI ontology that is allowed on either LegalResource or LegalExpression, the property is asserted on LegalResource. If not, a warning is displayed : *“Although officially allowed, finding this property elsewhere than on LegalResource is suspicious”*. Experience has shown that this is a typical inconsistency found in the way ELI metadata is published.
- a rule that checks that every property used in the ELI namespace is actually declared in the ELI OWL file
- some rules to verify that the values of some properties belongs to a recommended vocabulary:
 - for `eli:language`, the language authority table of the EU
 - for `eli:relevant_for` and `eli:jurisdiction`, the ATU table of the EU
 - for `eli:is_about`, the Eurovoc thesaurus
 - for `eli:format`, a list of known, typical media-types
- It also checks that at least one date is available on legal resources : at least one `eli:date_document`, `eli:date_publication`, `eli:version_date`

Using the ELI validation rules in a SHACL validator

In order to use the ELI validation rules in a SHACL validator, the input data to be passed for validation must always contain the ELI OWL ontology. In other words, you should pass for validation a graph containing the actual data to be validated + the ELI OWL itself. This is necessary for:

- Node Shapes targets to resolve properly on subclasses (in core rules)
- Check the presence of each property in the OWL ontology (recommended rules)

In terms of rules, the system can either use:

1. only the core rules : strict validation of a webpage content
2. the core rules + recommended rules : validation of webpage content against strict and recommended rules
3. the core rules + ranges rules : validation of a complete ELI knowledge graph, strict

4. the core rules + ranges rules + recommended rules : validation of a complete ELI knowledge graph, recommended

Maintenance of the ELI validation rules

Rule files

- The ELI validation rules are maintained in Excel tables that follow the [syntax of the xls2rdf converter](#). They are then converted to RDF using this converter. The converter can be used in its online version but can also be used locally in order to be independent of the availability of the online converter.
- The rules are maintained in 3 separate files : core, ranges, and recommended.
- The 3 files are implemented for each version of the ontology, 1.3, 1.4, 1.5, hence a total of 9 files.
- In each file, the new/modified rules compared to the previous version are highlighted in yellow.
- Both the Excel files and the result of the conversion in SHACL RDF are provided as deliverables

Converting Excel files to SHACL

The spreadsheets are originally maintained online using Google spreadsheets. The original link to the online version of the spreadsheet can be found in the “Links” tab of each file. All online version are publicly accessible in read-only.

To convert one spreadsheet in SHACL, either:

1. Use the [online version of the xls2rdf converter](#)
2. Use the [command-line version of the converter](#)
3. Modify the original Google spreadsheet online (but you need to have the rights for this), and click on the “Link to SHACL” link in the “Links” tab